

Validity Analysis: MILU

Target context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Overall risk: **HIGH**

Deployment Context

Use case: Deployment scenario: A Hindi-speaking student in India is testing his/her preparation for competitive job-related examinations, with an AI system providing feedback on their responses. The goal is to evaluate the applicability of the benchmark and the quality of the LLM's feedback.

Domain: Educational assessment

Setting: Mobile application / enterprise software

Note: The deployment hypothesis should be tested using Hindi-language sentences from the benchmark, as I am familiar with the language and can provide evaluation in a subsequent stage.

Target population: A graduate student in North India interacting in the Hindi language. The student has completed their education primarily in Hindi, with limited exposure to English.

Dimension Scores

Dimension	Score	Rating	Confidence
Input Ontology ✓	3	Moderate gaps	high
Input Content △	2	Significant gaps	high
Input Form ✓	4	Minor gaps	high
Output Ontology △	1	Serious concern	high
Output Content	2	Significant gaps	medium
Output Form △	1	Serious concern	high
Average	2.2		

△ = highest concern ✓ = strongest dimension

Dimension Key

Abbr.	Dimension	Definition
IO	Input Ontology	Whether the benchmark's test case categories cover the query types expected in deployment.
IC	Input Content	Whether individual datapoint content is culturally and contextually appropriate for the target region.
IF	Input Form	Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.
OO	Output Ontology	Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.
OC	Output Content	Whether ground-truth labels align with the judgments of regional stakeholders.
OF	Output Form	Whether the expected output modality matches regional deployment needs and accessibility.

Overall Summary

MILU is the most relevant publicly available benchmark for Indic-language knowledge evaluation and provides genuine value for a portion of this deployment's needs — particularly for measuring MCQ-level accuracy on Indian Polity, History, Geography,

and Economics in Hindi. Its India-first design [Q11], real-exam sourcing [Q24, Q25], and 41-subject taxonomy [Q42] are well-suited to a central-exam preparation context in principle. However, two structural issues sharply limit its validity for THIS deployment: (1) Output form/ontology mismatch (Output Ontology=1, Output Form=1) — MILU evaluates only MCQ option selection, while the deployment requires open-ended Hindi explanatory rationales; this is a fundamental scope gap acknowledged by the paper itself [Q85] and confirmed by data inspection [DATASET-D1, D9]. (2) Hindi partition content concerns (Input Content=2) — dataset analysis found 100% is_translated=True in the 26-item Hindi validation sample [DATASET-D13], heavy Engineering/Tech skew misaligned with UPSC/SSC syllabi [D5, D6, D7, D16, D36], Current Affairs staleness [D4, D25, D33], and pervasive cross-language item duplication. Input form (4) is the strongest dimension; output ontology and output form are the weakest. The benchmark can serve as a partial accuracy probe for a subset of priority subjects but cannot validate the deployment's core output.

ID	Evidence
Q11	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q25	"We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages." (p.3)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D4	मई 2022 में इराक के सुलेमानिया में आयोजित तीरंदाजी एशिया कप 2022 स्टेज 2 अभियान में भारत ने कतिने स्वर्ण पदक जीते? ... आठ — Current Affairs question (2022) about archery medal count — time-bounded, may be stale
D5	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... मोटर की गतिबहुत अधिक होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus
D6	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: ... पल्सेटिंग डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS
D7	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। ... शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
D13	All 26 Hindi examples: is_translated = True — Every Hindi validation example in the sample is machine-translated from English
D16	फॉरट्रान 77 के फक्सिड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? ... एक जंप लेबल या फॉरमेट लेबल — Computer Science / Fortran programming question in Hindi — very niche for UPSC/SSC aspirants
D25	ढरहरि - भारु 2022 ढरिं रेरी रान ढरिणसडा चेढां ढरिं, आढ आढमी ढरिटी (AAP) ढे ... ढरिणस — Current Affairs (2022 state elections) in Punjabi — politically specific, time-bounded
D33	फवेरुयारि- मारु २०२२ ए अनुष्ठरि राजुय बरिणसडा नरिबानन, आम आढमि ढरिटी (AAP) ... ढरिणस — 2022 state elections Current Affairs — AAP, UP, Goa all named; time-bounded content
D36	एएम की बैडवडिथ _____ है। ... 1110 kHz — AM bandwidth engineering question — highly technical, `is_translated: True`; unlikely in UPSC GS

Practical Guidance

What This Benchmark Measures

MILU measures MCQ-label accuracy on Hindi-language Indian-domain knowledge across 8 macro-domains and 41 subjects. For this deployment, it provides moderate evidence for a model's ability to produce correct verdicts on UPSC/SSC-style Polity, History, Geography, and Economics MCQs in Hindi (Input Ontology partial coverage, Input Form strong, Input Content partial). It offers no evidence on the deployment's core output: the Hindi explanatory rationale (Output Ontology and Output Form both

score 1).

Construct Depth

Construct depth is shallow for this deployment. MILU probes whether a model can rank the correct option among up to four — a necessary but far-from-sufficient signal for exam-feedback quality. It does not probe pedagogical clarity, factual rationale accuracy, fluency, register appropriateness, or robustness to authentic Hindi-medium exam phrasing. Combined with cross-language duplication, possible high translation rate in Hindi, and Current Affairs staleness, even the verdict-accuracy signal is partially construct-irrelevant for 2025–26 central-exam readiness.

What Else You Need

Three supplementations are essential: (1) A custom Hindi-language rationale-quality evaluation framework with rubric covering factual accuracy, fluency, pedagogical clarity, and register — no public resource exists [WEB-13, WEB-14]. (2) A central-exam-specific subset of MILU filtered to UPSC/SSC/banking items with explicit Mathematics/Reasoning and Current Affairs categories, ideally drawing on natively authored Hindi items rather than translated content. (3) A current-affairs evaluation set refreshed annually with 2024–2026 events to address staleness.

ID	Evidence
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)

Input Ontology (IO)

Score: 3/5 — Moderate gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Input Ontology definition: Whether the benchmark's test case categories cover the query types expected in deployment.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's 8 macro-domain / 41-subject taxonomy [Q42] aligns partially with central-exam priority subjects: Social Sciences and Law & Governance map to History/Polity, while Arts & Humanities covers cultural GK. However, the taxonomy does not separately enumerate Mathematics/Reasoning or Current Affairs as named categories — both heavily weighted in UPSC CSAT and SSC CGL Tier 1 [WEB-9, WEB-10, WEB-15]. Dataset analysis confirms reasoning and quantitative items exist (in the English partition: D8, D9), but the Hindi sample shows a heavy skew toward translated Engineering & Tech items (DC motor, half-wave rectifier, transformer, AM bandwidth, Fortran 77) that are peripheral to UPSC GS / SSC GA syllabi. State-level and central-level items are pooled without metadata distinguishing them [Q27], making it impossible to filter for central-exam scope [D24, D27].

ID	Evidence
Q27	"Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years." (p.3)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
D8	Fill in the gap below with suitable word. Fruit : Reap:: Flower : ____... Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
D24	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? ... Estimate Committee: All members of this committee are from Assembly only. — State-level (Maharashtra) governance question in English partition — not central exam scope
D27	मध्यप्रदेश में निम्नलिखित में से कौन सा उद्योग है? ... मारवा — State-level GK (Madhya Pradesh industrial geography) in Telugu — regional, not central exam
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)
WEB-10	SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension (https://www.pw.live/ssc/exams/ssc-cgl-syllabus)
WEB-15	UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) (https://vajiramandravi.com/upsc-exam/csatsyllabus/)

Strengths

- Explicit India-first design covering Science, Social Sciences, Humanities, Arts, Business Studies, Law [Q23, Q74] aligns with multi-domain GK breadth required for UPSC/SSC.
- Hierarchical 8-domain × 41-subject structure [Q42] supports per-subject performance analysis relevant to syllabus mapping.
- Reasoning and quantitative aptitude content is present in MILU's taxonomy, confirmed empirically in the English partition [DATASET-D8, D9].

ID	Evidence
----	----------

Q23	"This benchmark covers many domains, including Science, Social Sciences, Humanities, Arts, Business Studies, and Law, among others." (p.3)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q74	"In this paper, we introduced MILU—Multilingual Indic Language Understanding Benchmark—a comprehensive benchmark specifically designed to evaluate LLMs across 11 Indic languages, spanning diverse domains and culturally relevant subjects." (p.8)
D8	Fill in the gap below with suitable word. Fruit : Reap:: Flower : _____ Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition

Checklist

Checklist item definitions:

ID	Assessment Question
IO-1	Identify test case categories required for regional deployment
IO-2	Check if taxonomy omits regionally relevant categories
IO-3	Check if taxonomy includes irrelevant categories
IO-4	Document category gaps harming content validity

Checklist responses:

IO-1: For central-exam aspirants (UPSC/SSC/banking), priority categories are GK, Current Affairs, History, Polity, Mathematics/Reasoning, Hindi proficiency, Geography, Economics (elicitation Q1; [WEB-9, WEB-10, WEB-15]).

IO-2: Mathematics/Reasoning and Current Affairs are not named as MILU macro-domains [Q42]; they may be subsumed under other domains but lack explicit enumeration. Dataset findings confirm reasoning/quant items exist in the English partition [DATASET-D8, D9] but not whether the Hindi partition has comparable representation. State-level vs. central-exam labeling is absent [DATASET-D24, D27].

IO-3: MILU pools state-PSC and central-exam items [Q27, Q92] and the observed Hindi sample contains low-priority Engineering/Tech and niche CS items (Fortran 77, AM bandwidth) that are construct-irrelevant for UPSC/SSC GS [DATASET-D5, D6, D7, D16, D36].

IO-4: Two notable gaps: (a) absence of explicit Current Affairs and Mathematics/Reasoning macro-domains hampers syllabus-aligned scoring; (b) inability to separate central-exam from state-PSC items at the metadata level injects construct-irrelevant variance for this deployment.

ID	Evidence
Q1	"MILU spans 8 domains and 41 subjects across 11 Indic languages, reflecting both general and culturally specific knowledge." (p.1)
Q27	"Our benchmark includes questions from over 40 different types of exams conducted both at the national and state levels over recent years." (p.3)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q92	"We collected our questions from over 40 exam types ranging from various National and state level civil service examinations to examinations conducted by various government and private organizations." (p.17)
D5	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus
D6	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: ... प्लसेटिंग डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS
D7	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। ... शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep

D8	Fill in the gap below with suitable word. Fruit : Reap:: Flower : ____... Bloom — Logical Reasoning/analogy question in English — confirms presence of reasoning subject
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
D16	फॉरट्रान 77 के फकिस्ड फॉरमेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? ... एक जंप लेबल या फॉरमेट लेबल — Computer Science / Fortran programming question in Hindi — very niche for UPSC/SSC aspirants
D24	Which one of the following statements about Legislative Committee system of Maharashtra is not correct? ... Estimate Committee: All members of this committee are from Assembly only. — State-level (Maharashtra) governance question in English partition — not central exam scope
D27	మధ్యప్రదేశ్‌లోని బన్‌మూర్ హరికాండ అభివృద్ధి కేంద్రం ఏ జిల్లాలో ఉంది? ... మౌరసా — State-level GK (Madhya Pradesh industrial geography) in Telugu — regional, not central exam
D36	एएम की बैंडविड्थ _____ है। ... 1110 kHz — AM bandwidth engineering question — highly technical, 'is_translated: True'; unlikely in UPSC GS
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)
WEB-10	SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension (https://www.pw.live/ssc/exams/ssc-cgl-syllabus)
WEB-15	UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) (https://vajiramandravi.com/upsc-exam/csatsyllabus/)

Information Gaps

- Proportion of MILU Hindi items aligned to UPSC/SSC/banking vs. state PSC is not published.
- Whether Mathematics/Reasoning and Current Affairs are present in the Hindi partition at central-exam syllabus weight is not quantifiable from the small validation sample.

Requires Expert Verification

- A subject-matter expert in UPSC/SSC syllabus design should map MILU's 41 subjects to central-exam syllabus weights and quantify representation in the Hindi partition's test split.

Remediation

Gap 1: Mathematics/Reasoning and Current Affairs are not separately enumerated in MILU's 8 macro-domains; central-exam vs. state-PSC items are commingled without metadata.

Recommendation: Add metadata layer mapping each MILU item to (a) UPSC/SSC/banking syllabus category (including explicit Math/Reasoning, Current Affairs), (b) central-exam vs. state-PSC scope. This enables filtered evaluation aligned to syllabus weights [WEB-9, WEB-10, WEB-15].

ID	Evidence
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)
WEB-10	SSC CGL Tier 1 covers Reasoning, General Awareness, Quantitative Aptitude, English Comprehension (https://www.pw.live/ssc/exams/ssc-cgl-syllabus)
WEB-15	UPSC CSAT Paper 2 covers Quantitative Aptitude and Logical Reasoning (qualifying) (https://vajiramandravi.com/upsc-exam/csatsyllabus/)

Input Content (IC)

Score: 2/5 — Significant gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Input Content definition: Whether individual datapoint content is culturally and contextually appropriate for the target region.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU's stated design priorities — India-first sourcing, regional cultural content, sourcing from real exam papers **[Q11, Q12, Q24, Q25]** — are well aligned with the deployment's content needs. However, dataset analysis reveals two critical issues for the Hindi partition: (1) all 26 Hindi validation examples carry `is_translated=True` **[DATASET-D13]**, suggesting the Hindi partition may contain a much higher translated proportion than the documented overall ~25% **[Q47]**; (2) cross-language item duplication is pervasive **[D14, D15, D29]**, reducing effective Hindi-specific content. North India sub-regional content is sparse in the observed Hindi sample (only **DATASET-D3** on UP minerals). Current Affairs items are time-bounded to 2014–2022 events **[DATASET-D4, D25, D33]**, creating staleness risk for 2025–26 exam cycles. While translated items use fluent Devanagari with low code-mixing **[DATASET-D12, D21]**, the linguistic register may not match authentic Hindi-medium exam papers.

ID	Evidence
Q11	"We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams." (p.2)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q25	"We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages." (p.3)
Q47	"Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder" (p.4)
D3	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — State-specific GK question about Uttar Pradesh minerals — North India sub-regional content
D4	मई 2022 में इराक के सुलेमानिया में आयोजित तीरंदाजी एशिया कप 2022 स्टेज 2 अभियान में भारत ने कतिने स्वर्ण पदक जीते? ... आठ — Current Affairs question (2022) about archery medal count — time-bounded, may be stale
D12	रेडक्लफि रेखा भारत की सीमाओं को नमिनलखिति में से किस देश के साथ वभिाजति करती है? ... पाकस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D13	All 26 Hindi examples: <code>is_translated = True</code> — Every Hindi validation example in the sample is machine-translated from English
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूरगिर करने वाले वदिशी मुद्रा भंडार की हानिका तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but <code>'is_translated: True'</code> , confirms translation pipeline
D21	एक भौगोलिक क्षेत्र में नमिनलखिति वशिषिट वशिषिताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing
D25	ढरद्वरी - ढारुच 2022 दूँचि ठेही रान दपिनमडा चेठं दूँचि, आम आदमी पारटी (AAP) ने ... पंजाब — Current Affairs (2022 state elections) in Punjabi — politically specific, time-bounded
D29	ഭരണഘടനാ ഭേദഗതിയുടെ പേര് 'സാധാരണ കലാപം' എന്ന പദം ഭരണഘടനയിൽ ചേർക്കുന്നത് എപ്പോൾ? ... 44-ാമത് ഭരണഘടനാ ഭേദഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication
D33	ফেব্রুয়ারি- মার্চ ২০২২ এ অনুষ্ঠিত রাজ্য বহিানসভা নির্বাচনে, আম আদমি পার্টি (AAP) ... পাঞ্জাব — 2022 state elections Current Affairs — AAP, UP, Goa all named; time-bounded content

Strengths

- Source content is real Indian competitive exam papers [Q24, Q25], grounding the input in authentic exam ecosystems.
- India-first design explicitly includes culturally relevant subjects — local history, arts, festivals, laws [Q12, Q2].
- Constitutional law, Mughal/medieval North Indian history, and Indian GK are present in Hindi [DATASET-D1, D2, D3, D11, D18] — directly relevant to UPSC/SSC content.
- Observed Hindi items use clean Devanagari with minimal English code-mixing [DATASET-D12, D21], respecting the deployment's ~10% ceiling.

ID	Evidence
Q2	"With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science." (p.1)
Q12	"These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science." (p.2)
Q24	"These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others." (p.3)
Q25	"We gathered exam-specific questions by scraping various online exam portals that offer previously released question papers from various exams in multiple different languages." (p.3)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D2	मुगल ग्रंथ 'मासिर-ए-आलमगीरी' के रचयिता कौन हैं? ... साकी मुस्तईद खान — Mughal-period history question in Hindi — directly relevant to North India medieval history
D3	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — State-specific GK question about Uttar Pradesh minerals — North India sub-regional content
D11	'मानसून' शब्द की उत्पत्तिकिस भाषा से हुई है? ... अरबी भाषा — General Knowledge question in Hindi (monsoon etymology) — accessible, pan-India GK
D12	रेडक्लिफ रेखा भारत की सीमाओं को नमिनलखिति में से किस देश के साथ विभाजित करती है? ... पाकिस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D18	नमिनलखिति में से कौन एक प्रसिद्ध भारतीय चित्रकार थीं, जिन्हें कभी-कभी भारत की फर्दि काहलो के नाम से जाना जाता है? ... अमृता शेरगलि — Arts & Culture GK in Hindi — pan-India Indian art figure
D21	एक भौगोलिक क्षेत्र में नमिनलखिति विशेषित विशेषताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing

Checklist

Checklist item definitions:

ID	Assessment Question
IC-1	Do inputs require region-specific cultural/geographic/dialectal knowledge?
IC-2	Does culturally sensitive content align with target culture?
IC-3	Flag inputs assuming Western-specific knowledge
IC-4	Regional annotator recruitment for culturally sensitive instances
IC-5	Document content issues harming content validity

Checklist responses:

IC-1: Yes, deployment requires both pan-India GK and North India sub-regional knowledge [elicitation Q2]. MILU partially addresses this — pan-India GK is well represented [DATASET-D1, D11, D18, D22], but North India sub-regional content (Chhath Puja, zamindari, North Indian freedom fighters) is sparse in observed Hindi sample (only DATASET-D3).

IC-2: Cultural alignment is partially achieved through India-first sourcing [Q11, Q24], but the dominant translation path for the Hindi partition (DATASET-D13: 100% is translated in sample) means linguistic-cultural register may not match Hindi-medium

exam conventions.

IC-3: No Western-specific content was observed; the benchmark is India-centric. However, translated engineering items [DATASET-D5, D6, D7, D36] inject construct-irrelevant content that mismatches central-exam syllabus.

IC-4: INSUFFICIENT DOCUMENTATION — paper does not report whether annotators or reviewers were Hindi-medium graduates or domain experts familiar with central-exam content [Q87 mentions only 'AI4Bharat volunteers'].

IC-5: Critical content issues: (a) probable over-representation of translated items in Hindi partition; (b) Current Affairs staleness (events 2014–2022 in a benchmark to be used for 2025–26 prep); (c) cross-language duplication [D14, D15, D29] reducing genuine Hindi-source diversity; (d) sparse North India sub-regional items observed.

ID	Evidence
Q2	“With an India-centric design, MILU incorporates material from regional and state-level examinations, covering topics such as local history, arts, festivals, and laws, alongside standard subjects like science.” (p.1)
Q11	“We designed MILU with an India-first perspective by collecting questions from various national, state, and regional exams.” (p.2)
Q12	“These questions include culturally relevant subjects such as local history, arts, festivals, and laws, alongside traditional academic subjects like science.” (p.2)
Q24	“These questions were sourced following an approach similar to AGIEVAL (Zhong et al., 2023), collecting the questions from various public exams taken by individuals intending to either pursue higher studies or seek career advancements, such as qualification tests and national and state-level civil services exams, among others.” (p.3)
Q44	“For subjects with insufficient questions, we sampled questions from the English set from that subject and translated them into the required language using GPT-4O.” (p.4)
Q47	“Table 2 shows the overall statistics of MILU. Of the total 79K questions, only 25% of questions are translated from English, with the remainder” (p.4)
Q87	“We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits.” (p.9)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D3	नमिनलखिति में से कौन सा खनजि उत्तर प्रदेश में नहीं पाया जाता है? ... अभ्रक — State-specific GK question about Uttar Pradesh minerals — North India sub-regional content
D5	जब एक डीसी सीरीज मोटर बर्ना लोड के चलती है: ... मोटर की गतिबहुत अधिक होती है — Engineering/Physics question in Hindi, `is_translated: True` — not in UPSC/SSC core syllabus
D6	हाफ वेव रेक्टिफायर का आउटपुट क्या होता है: ... पल्सेटिंग डीसी — Technical engineering question in Hindi, `is_translated: True` — domain not central to UPSC/SSC GS
D7	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। ... शेल — Electrical engineering, translated — technical domain not prioritized for Hindi-medium UPSC prep
D11	'मानसून' शब्द की उत्पत्तिकिस भाषा से हुई है? ... अरबी भाषा — General Knowledge question in Hindi (monsoon etymology) — accessible, pan-India GK
D13	All 26 Hindi examples: is_translated = True — Every Hindi validation example in the sample is machine-translated from English
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को द्रगिर करने वाले वदिशी मुद्रा भंडार की हानिका तात्कालकि कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline
D18	नमिनलखिति में से कौन एक प्रसदिध भारतीय चतिर्कार थीं, जनिहे कभी-कभी भारत की फ्रदि काहलो के नाम से जाना जाता है? ... अमृता शेरगलि — Arts & Culture GK in Hindi — pan-India Indian art figure
D22	What was the first capital of the Bahamani Kingdom? ... Gulbarga — India-specific historical knowledge (Deccan Sultanate) — same question appears across all languages
D29	ഭരണീയ അടിയന്തരാവസ്ഥ പ്രഖ്യാപിക്കാൻ 'സായുധ കലാപം' എന്ന്ന പദം ഭരണഘടനയിൽ ചരൽത്തൽ എപ്പരൾ? ... 44-ാമൽ ഭരണഘടനാ ഭരഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication
D36	एएम की बैडवडिथ _____ है। ... 1110 kHz — AM bandwidth engineering question — highly technical, `is_translated: True`; unlikely in UPSC GS
WEB-9	Current Affairs is heavily and variably weighted in UPSC GS Paper 1, raising staleness concerns for older exam scrapes (https://testbook.com/ias-preparation/upsc-prelims-subject-wise-weightage)

Information Gaps

- True translation rate in the Hindi test split (vs. observed 100% in 26-item validation sample) is unknown.
- Proportion of MILU Hindi items aligned with central exams vs. state PSCs is undocumented.
- Item-level North India sub-regional content breakdown is not published.
- Annotator demographics — whether reviewers were Hindi-medium graduates familiar with the target student linguistic register — not reported.

Requires Expert Verification

- A Hindi-medium UPSC/SSC subject-matter expert should review a stratified sample of MILU Hindi test items for: (a) authenticity of register vs. authentic Hindi-medium exam papers, (b) translation artifact rate, (c) syllabus alignment with central exams, (d) North India sub-regional content presence.

Remediation

Gap 1: Hindi partition appears to contain a high proportion of translated items (DATASET-D13: 100% in observed sample), pervasive cross-language duplication, and stale Current Affairs.

Recommendation: Curate a deployment-specific Hindi subset: (a) prefer items with is_translated=False where they exist; (b) deduplicate across languages; (c) replace pre-2023 Current Affairs items with a fresh annual current-affairs probe; (d) ensure North India sub-regional items (Chhath Puja, zamindari, North Indian freedom fighters) are sufficiently represented via custom item authoring.

Gap 2: North India sub-regional content (Chhath Puja, zamindari, regional freedom fighters) is sparsely observed in the Hindi sample (1 of 26 items).

Recommendation: Author or source ~200–500 additional Hindi items specifically covering North India sub-regional content within central-exam framing, drawing from authentic UPSC/SSC previous-year papers.

ID	Evidence
D13	All 26 Hindi examples: is_translated = True — Every Hindi validation example in the sample is machine-translated from English

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

MILU is text-only with explicit exclusion of reading-comprehension passages, image-based items, and >4-option items [Q33], plus INDICLID + Unicode-based language filtering [Q34], duplicate removal [Q35], and multi-stage manual cleaning [Q31, Q32, Q36]. This matches the deployment’s text-only mobile interface and standard Devanagari script with no RTL/capitalization concerns. Observed Hindi items render in standard Devanagari with minimal Roman intrusion [DATASET-D12, D21]. One residual concern: dataset analysis observed at least two malformed items with truncated stems across language partitions [DATASET-D34, D35], suggesting filtering imperfections.

ID	Evidence
Q31	“Despite our best efforts to maintain the quality of questions collected, some amount of noise or errors may still be present. To address potential noise in the questions, we employ multiple layers of manual and automated cleaning filters.” (p.4)
Q32	“Initially, we manually review a large sample of questions to detect and eliminate potential sources of noise.” (p.4)
Q33	“During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency.” (p.4)
Q34	“To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language.” (p.4)
Q35	“To further refine the dataset, we remove any duplicate questions to retain only the unique ones.” (p.4)
Q36	“As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors.” (p.4)
D12	रेडक्लफि रेखा भारत की सीमाओं को नमिन्लखिति में से कसि देश के साथ वभिजति करती है? ... पाकस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D21	एक भौगोलिक क्षेत्र में नमिन्लखिति वशिष्ट वशिषताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — “brought a” with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിൽപ്പം — Incomplete stem (“Directions: marble can be used”) — same truncated structure appears across languages

Strengths

- Text-only MCQ format directly matches deployment input modality.
- Devanagari script handling is well-supported [Q34], no signal-distribution mismatch for Hindi.
- Multi-layer cleaning pipeline [Q31, Q32, Q36] reduces noise.
- Observed code-mixing in Hindi items remains below the ~10% ceiling [DATASET-D12, D21].

ID	Evidence
Q31	“Despite our best efforts to maintain the quality of questions collected, some amount of noise or errors may still be present. To address potential noise in the questions, we employ multiple layers of manual and automated cleaning filters.” (p.4)
Q32	“Initially, we manually review a large sample of questions to detect and eliminate potential sources of noise.” (p.4)

Q34	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q36	"As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors." (p.4)
D12	रेडक्लफि रेखा भारत की सीमाओं को नमिनलखिति में से कसि देश के साथ वभिजति करती है? ... पाकस्तान — Geography GK question in Hindi — relevant to UPSC/SSC syllabus, Devanagari script, no English terms
D21	एक भौगोलिक क्षेत्र में नमिनलखिति वशिष्ट वशिषताएँ हैं ... सदाबहार ओक के पेड़ ... भूमध्यसागरीय — Geography question (Mediterranean climate) — translated, pan-India geography, no English code-mixing

Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution matches: text-only Devanagari [Q33, Q34] aligns with the deployment's mobile text interface. No image/audio modality mismatch.

IF-2: Regional infrastructure (Android-dominant smartphones, Devanagari Unicode standard) supports text-only MCQ ingestion. Bandwidth-sensitive but format-compatible [WEB-6].

IF-3: Domain-specific form: MCQ-with-4-options matches central-exam Prelims pattern (UPSC, SSC, banking are MCQ-based). Reading-comprehension exclusion [Q33] removes one common exam format but is acceptable for the deployment's objective-prep scope.

IF-4: Minor form mismatches: (a) some items have truncated stems suggesting reading-comprehension remnants [DATASET-D34, D35]; (b) reading-comprehension exclusion means RC-style UPSC CSAT questions are not represented.

ID	Evidence
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q34	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നത് ... ശിൽപ്പം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages
WEB-6	TRAJ data shows internet penetration patchy in Bihar — relevant to deployment connectivity assumptions (https://trai.gov.in/sites/default/files/2024-11/Cons_P_14092023.pdf)

Information Gaps

- Frequency of malformed/truncated items in the full Hindi test split is unknown.

Output Ontology (OO)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Output Ontology definition: Whether the benchmark's output categories and scoring criteria reflect regionally valid decision boundaries.

Each section below is followed by an evidence table. Three types of evidence are cited: [QN] — verbatim quotes extracted from the benchmark paper; [WEB-N] — web sources consulted during assessment; [DN] / [DATASET-DN] — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

This is a fundamental mismatch. MILU's output ontology is a single MCQ label — one of up to four options [Q33, Q55] — and the dataset schema confirms only a 'target' field with no rationale or explanation field [DATASET-D1, D9]. The deployment requires a correct/incorrect verdict PLUS a substantive Hindi-language explanatory rationale [elicitation Q3]. MILU has no rubric, no label set, and no scoring function for rationale quality. The 41-subject taxonomy [Q42] supports per-subject accuracy reporting but cannot validate explanation quality. Compounding this, the benchmark itself reports models perform worst in Arts & Humanities, Social Sciences, and Law & Governance [Q6, Q20, Q62, Q71, Q77] — precisely the deployment's priority subjects (History, Polity, GK).

ID	Evidence
Q3	"We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.1)
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q20	"Our domain-wise analysis reveals that models perform poorly in culturally relevant areas, such as Arts & Humanities and Social Sciences, compared to more general fields like STEM." (p.2)
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q55	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q62	"When analyzed across domains, the models generally perform worse in Arts, Humanities, and Social Sciences than in STEM subjects." (p.6)
Q71	"We analyze the performance of various base and instruct models across multiple domains and languages. Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
Q77	"Additionally, the domain-specific analysis indicates that models perform better in general fields such as STEM while facing challenges in culturally relevant subjects like Arts, Humanities, and Law, highlighting the lack of this knowledge in the current models and datasets." (p.8)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition

Strengths

- 41-subject × 8-domain hierarchical taxonomy [Q42] supports per-subject diagnostic accuracy reporting, useful for identifying weak areas.
- Closed-form MCQ accuracy is unambiguous for the verdict portion of the deployment's required output.

ID	Evidence
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)

Checklist

Checklist item definitions:

ID	Assessment Question
OO-1	Review output label categories for regional relevance
OO-2	Identify missing regionally specific categories
OO-3	Flag categories encoding non-regional values
OO-4	Consider stakeholder-driven taxonomy redesign
OO-5	Document taxonomy issues harming structural/content validity

Checklist responses:

OO-1: MCQ option labels are regionally relevant only insofar as the questions are India-sourced [[Q26](#)]. The output category space is, however, structurally incomplete for this deployment.

OO-2: Critical missing categories: rationale-quality dimensions (factual accuracy, fluency, pedagogical clarity, register appropriateness — listed in elicitation pedagogical_context). MILU has none of these.

OO-3: MCQ-only ontology does not encode non-regional values per se, but encodes an evaluation philosophy (option-selection only) that is misaligned with pedagogical-feedback deployments.

OO-4: Significant taxonomy redesign is required: a separate rationale-quality rubric must be developed, as MILU provides no foundation. The 2025 Hindi LLM benchmarking preprint [[WEB-14](#)] confirms no such Hindi rubric exists publicly.

OO-5: MILU's MCQ-only output ontology violates structural validity for this deployment (the construct of pedagogical feedback has internal structure — verdict + rationale — that MILU cannot represent) and content validity (rationale-quality categories are entirely absent).

ID	Evidence
Q6	"Domain-wise analysis indicates that models perform poorly in culturally relevant areas like Arts & Humanities and Law & Governance compared to general fields like STEM." (p.1)
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q42	"Following the manual merging of related clusters, we determine 41 distinct subject names, which fall into eight main domains: Arts and Humanities, Social Sciences, Environmental Sciences, Law and Governance, Health and Medicine, Science, Engineering and Technology, and Business Studies." (p.4)
Q55	"For multiple choice questions, given k possible answer strings, we select the answer string (a _i) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i \cdot))$ " (p.5)
Q71	"We analyze the performance of various base and instruct models across multiple domains and languages. Similar trends to those in Section (§5.2) are observed where the open models perform poorly in domains specific to Indian culture—such as Arts & Humanities, Social Sciences, and Law & Governance—but demonstrate higher performance in STEM fields." (p.7)
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)

Requires Expert Verification

- A pedagogical-NLP expert should design a Hindi exam rationale rubric covering factual accuracy, fluency, and pedagogical clarity — this is wholly outside MILU's scope.
-

Remediation

Gap 1: No taxonomy or label set for rationale quality dimensions exists in MILU.

Recommendation: Adopt or develop a multi-axis rationale-quality taxonomy (e.g., 4-axis: factuality, fluency, pedagogical helpfulness, Manak Hindi register), informed by Hindi-medium coaching pedagogy conventions. Expert verification by Hindi-medium UPSC/SSC educators is required.

Output Content (OC)

Score: 2/5 — Significant gaps | Confidence: medium

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Output Content definition: Whether ground-truth labels align with the judgments of regional stakeholders.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

For the MCQ-label correctness narrowly, MILU has reasonable provenance: answers are inherited from online exam portals where 'subject experts ensure the accuracy of the answers' **[Q26]**, with AI4Bharat volunteer manual audits **[Q87]** and researcher review of subject-tag clusters **[Q41]**. However, the paper reports no annotator demographics, no inter-annotator agreement statistics, and no formal QA protocol beyond manual sampling. Critically, the deployment's core output (Hindi rationale) has no ground-truth labels in MILU at all. Dataset analysis surfaces additional concerns: cross-language item duplication **[D14, D15, D29]** and malformed items with truncated stems **[DATASET-D34, D35]** suggest QA imperfections at the item level. ~55% of subject labels were assigned via machine translation + GPT-4O-MINI **[Q37, Q38]**, an automated process that may not preserve Hindi-specific terminology fidelity.

ID	Evidence
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q37	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q38	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूर करने के लिए विदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline
D29	ഭരണീയ അടിയന്തരാവസ്ഥ പര്യാപ്തമാക്കാനു് 'സാധുവ കലാപം' എന്ന്ന പദം ഭരണഘടനയിൽ ചരേത്തത് എപ്പാശ്? ... 44-ാമത് ഭരണഘടനാ ഭരേഗതി നിയമം വഴി — Same Constitutional Amendment question appears in Hindi (D1), Tamil, and Malayalam — cross-language duplication
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നതാണ് ... ശിൽപ്പം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages

Strengths

- Answer keys derive from established Indian competitive-exam ecosystems where subject experts validate correctness at source **[Q26]**.
- Multi-stage manual review applied to subject-tag clusters **[Q41]** and a manual audit pass **[Q87]**.
- Constitutional, historical, and GK answer labels in the observed Hindi sample appear factually correct **[DATASET-D1, D2, D22]**.

ID	Evidence
----	----------

Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)
Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D2	मुगल ग्रंथ 'मासिर-ए-आलमगीरी' के रचयिता कौन हैं? ... साकी मुस्तईद खान — Mughal-period history question in Hindi — directly relevant to North India medieval history
D22	What was the first capital of the Bahamani Kingdom? ... Gulbarga — India-specific historical knowledge (Deccan Sultanate) — same question appears across all languages

Checklist

Checklist item definitions:

ID	Assessment Question
OC-1	Do ground-truth labels reflect regional stakeholder perspectives?
OC-2	Assess potential annotator-regional population disagreement
OC-3	Review annotator demographics from documentation
OC-4	Consider label re-annotation by regional annotators
OC-5	Review aggregation methods for minority perspective erasure
OC-6	Document label issues harming convergent/external validity

Checklist responses:

OC-1: MCQ answer labels reflect Indian exam-portal subject-expert consensus [Q26], which is reasonable for central-exam content. However, regional stakeholder perspectives on rationale quality have no representation since rationale labels do not exist.

OC-2: INSUFFICIENT DOCUMENTATION on annotator-population disagreement risk for MCQ labels; annotator demographics are not reported beyond 'AI4Bharat volunteers' [Q87].

OC-3: Paper does not provide a Datasheet or Data Statement with annotator demographics — gap for assessing convergent validity.

OC-4: Re-annotation by Hindi-medium subject experts would be needed to (a) validate translation quality on the ~25%+ translated Hindi subset, (b) flag items where the answer is region-sensitive, (c) generate ground-truth Hindi rationales.

OC-5: INSUFFICIENT DOCUMENTATION on aggregation methods; the paper describes manual cluster review [Q41] but not how disagreements were resolved.

OC-6: Label issues: (a) no IAA statistics; (b) ~55% of topic labels assigned by automated MT+GPT pipeline [Q37, Q38]; (c) cross-language duplicates introduce identical labels across translated copies [D14, D15]; (d) malformed items have labels whose validity is questionable [DATASET-D34, D35]; (e) rationale ground-truth wholly absent.

ID	Evidence
Q26	"These portals typically tag questions manually with topic names and language details, and subject experts ensure the accuracy of the answers." (p.3)
Q37	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q38	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
Q41	"We manually review these clusters and assign appropriate subject labels." (p.4)

Q87	"We are also immensely grateful to the volunteers from the AI4Bharat team for their motivation and meticulous efforts in conducting manual audits." (p.9)
D14	What was the immediate cause for loss of foreign reserves triggering the financial crisis in 1991? ... Sharp rise in value of imports of oil & petroleum products — English original; same question appears in Hindi (D15) and Bengali — confirms translation pipeline
D15	1991 में वित्तीय संकट को दूरगिर करने वाले वदेशी मुद्रा भंडार की हानि का तात्कालिक कारण क्या था? ... तेल और पेट्रोलियम उत्पादों के आयात के मूल्य में तीव्र वृद्धि — Hindi translation of D14 — fluent Hindi, but `is_translated: True`, confirms translation pipeline
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നതാണ് ... ശിരസ്സം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages

Information Gaps

- Annotator demographics, IAA statistics, and disagreement-resolution protocols are not reported.
- Translation-quality verification for the Hindi partition is not documented.
- No Hindi rationale ground truth exists in the benchmark.

Requires Expert Verification

- A Hindi-medium subject-matter expert should audit a representative Hindi sample for label correctness, especially for translated items and items with stems that may have been corrupted by RC-passage stripping.

Remediation

Gap 1: No annotator demographics, IAA statistics, or formal QA protocol reported; ~55% of topic labels assigned by automated GPT-4O-MINI [Q37, Q38]; malformed items observed [DATASET-D34, D35].

Recommendation: Conduct a Hindi-medium expert audit on a stratified sample of the Hindi test split: verify answer correctness, translation-register quality, item completeness, and subject-tag accuracy. Report IAA. Filter or repair malformed items prior to deployment-grade evaluation.

ID	Evidence
Q37	"Upon examination, we found that approximately 45% of questions were accurately labeled with a topic name, while the remaining questions lacked this information." (p.4)
Q38	"To address this issue, we first translate the untagged questions into English using INDICTRANS2 (Gala et al., 2023) and then prompt GPT-4O-MINI model to assign an appropriate topic name to the question." (p.4)
D34	The owner of the textile shop brought a ... Calculator — Incomplete stem — "brought a" with no further context; question appears malformed
D35	ദിശകൾ: മാർബിൾ ഉപയോഗിക്കാവുന്നതാണ് ... ശിരസ്സം — Incomplete stem ("Directions: marble can be used") — same truncated structure appears across languages

Output Form (OF)

Score: 1/5 — Serious concern | Confidence: high

Benchmark: MILU | Context: Hindi-Medium Competitive Exam Aspirants — North India (Central Exams)

Output Form definition: Whether the expected output modality matches regional deployment needs and accessibility.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

The output form mismatch is fundamental and acknowledged by the paper itself. MILU evaluates via log-likelihood option selection [\[Q52, Q53, Q54, Q55\]](#) or zero-shot JSON-formatted option choice for API models [\[Q56, Q57\]](#). The authors note explicitly that this 'may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting' [\[Q85\]](#). The deployment requires open-ended Hindi-language explanatory generation — a fundamentally different output form. Dataset analysis confirms the schema contains only a 'target' MCQ label field [\[DATASET-D1, D9\]](#) with no infrastructure for evaluating fluency, correctness, or pedagogical quality of free-text Hindi output. External evidence confirms no comparable Hindi-language NLG benchmark for exam rationale exists [\[WEB-13, WEB-14\]](#).

ID	Evidence
Q52	"For non-API-based models, we use the LM-EVALUATION-HARNESS (Gao et al., 2024; Biderman et al., 2024) to ensure clean and reproducible evaluations." (p.5)
Q53	"We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020)." (p.5)
Q54	"Specifically, the log-likelihood of an answer (a) given the question (x), i.e., $\log P(a)$ " (p.5)
Q55	"For multiple choice questions, given k possible answer strings, we select the answer string (a _i) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q56	"The API-based models are evaluated using the generative approach due to the lack of support for prompt log probabilities." (p.5)
Q57	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
D1	राष्ट्रीय आपातकाल घोषित करने के लिए 'सशस्त्र विद्रोह' शब्द संविधान में कब जोड़ा गया? ... 44वें संविधान संशोधन अधिनियम द्वारा — Polity/Constitution question in Hindi, about 44th Constitutional Amendment — UPSC core topic
D9	If Rs. 4000 becomes Rs. 5760 in 2 years at compound interest ... what is the annual rate of interest? ... 20 percent — Mathematics/Quantitative Aptitude question (compound interest) — confirms presence in English partition
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)

Strengths

- MCQ accuracy can validate the verdict-correctness portion of the deployment's output.
- Per-language, per-subject accuracy reporting [\[Q59, Q67\]](#) supports diagnostic identification of weak areas.
- Multiple shot configurations (0/1/5-shot) [\[Q50\]](#) provide robustness checks on the verdict accuracy.

ID	Evidence
Q50	"All models, except for proprietary models and LLAMA-3.1-405B, are tested under 0-shot, 1-shot, and 5-shot setups." (p.5)
Q59	"We report 5-shot accuracies for all open models (except for LLAMA-3.1-70B-INSTRUCT & LLAMA-3.1-405B for which we report 0-shot accuracy) with the accuracy averaged across all the domains per language." (p.6)

Q67	"Table 4: Evaluation results of all the language-specific fine-tuned models on MILU. We report domain level 5-shot accuracies for all the models on the language supported by the model. Higher values indicate better model performance for the given domain." (p.7)
-----	---

Checklist

Checklist item definitions:

ID	Assessment Question
OF-1	Does output modality match regional deployment needs?
OF-2	Assess text-to-speech availability for speech output
OF-3	Consider literacy rates and accessibility requirements
OF-4	Document form mismatches harming external validity

Checklist responses:

OF-1: Major mismatch: MILU outputs MCQ option labels [Q55, Q57]; deployment requires open-ended Hindi rationale plus verdict [elicitation Q3]. The verdict portion is partially covered; the rationale portion is wholly uncovered.

OF-2: Not applicable — deployment is text-only, no TTS requirement.

OF-3: Target population is graduate-level (high literacy within cohort), so accessibility is not the bottleneck — the bottleneck is rationale quality, which MILU cannot evaluate.

OF-4: Form mismatches: (a) MCQ vs. open-ended generation; (b) log-likelihood scoring [Q53] vs. free-text Hindi generation quality; (c) no fluency/coherence/pedagogical metrics. The paper itself flags this limitation [Q85].

ID	Evidence
Q3	"We evaluate over 42 LLMs, and find that current LLMs struggle with MILU, with GPT-4o achieving the highest average accuracy at 74%." (p.1)
Q53	"We use the log-likelihood method, where the probability of a given output string is computed by conditioning it on some provided input (Brown et al., 2020)." (p.5)
Q55	"For multiple choice questions, given k possible answer strings, we select the answer string (ai) with the highest conditional log probability, i.e., $\text{argmax}(\log P(a_i))$ " (p.5)
Q57	"We explicitly prompt these models to generate the correct response in a structured JSON format to simplify response parsing." (p.5)
Q85	"Finally, our evaluation primarily relies on the log-likelihood approach, which may yield different results compared to other established methods, such as generation-based evaluation and chain-of-thought (CoT) prompting." (p.9)
WEB-13	IndicGenBench (ACL 2024) covers Hindi generation but not exam rationale quality (https://aclanthology.org/2024.acl-long.595/)
WEB-14	arXiv:2508.19831 confirms Hindi instruction-following / conversational benchmarks 'largely unavailable publicly' (https://arxiv.org/html/2508.19831v1)

Requires Expert Verification

- A Hindi-language NLG evaluation specialist should design a custom rationale-quality rubric (factual accuracy, fluency, pedagogical clarity, register) since no off-the-shelf Hindi resource exists [WEB-13, WEB-14].

Remediation

Gap 1: MILU evaluates only MCQ option selection; deployment requires open-ended Hindi rationale generation.

Recommendation: Design and operationalize a separate Hindi rationale-quality benchmark with (a) human-rated rubric (factual accuracy, fluency, pedagogical clarity, register), (b) ~500–1000 expert-authored gold rationales for a stratified sample of MILU Hindi items, (c) automated metrics (BLEURT/COMET-style for Hindi, plus LLM-as-judge calibrated against expert ratings).